HearSense

AI hearing screening & monitoring

iQOO
HEALTHTECH
Round-1 Brief

■ **SENSOR:** Speaker (Web Audio tones) · Microphone (room-noise gate)

The problem

Hearing loss is hugely under-detected; people wait ~7 years before testing. Clinical pure-tone audiometry needs a soundproof booth, calibrated gear and an audiologist — out of reach for most, especially in rural areas.

How it works

- Mic checks the room is quiet enough (the QC most apps skip).
- Digits-in-noise: repeat spoken digit triplets mixed with noise, one ear at a time.
- An adaptive staircase homes in on your signal-to-noise threshold (dB SNR).
- Optional pure-tone sweep (500 Hz–8 kHz) renders an audiogram-style profile.
- On-device AI explains the result, tracks change, and advises referral.

Example output

> Speech-in-noise score: –6 dB SNR — within normal range
>
> Right ear: likely mild loss in high frequencies (2–4 kHz)
>
> High-frequency hearing dropped vs last month — see an audiologist
>
> Room too noisy (58 dB) — move somewhere quieter

Metrics

Hearing score · speech-in-noise (dB SNR) · high-freq threshold · category (Normal→Refer)

Novelty & positioning

Medium — leads with calibration-free digits-in-noise (WHO hearWHO method); differentiator is longitudinal monitoring + regional digit sets.

Target users

Adults 40+ · noise-exposed workers · rural/low-resource populations · anyone monitoring hearing

Assistive / 'cure' feature

Prevention & protection guidance (60/60 rule, ear protection); routes to audiologist for hearing aids / treatment.

Research backing

- Validated smartphone hearing screens: 75–100% sens/spec (MDPI 2025 review)
- Digits-in-noise needs no calibration — why WHO adopted it (PMC 2019)
- DIN ≈90% correlation with pure-tone audiometry, ~3 min (MDPI 2024)

On-device AI coach	**Personal baseline + drift**	**History & charts**	**Monthly PDF report**	**Contact-a-speciali st**	**100% private / offline**

Part of the **SenseKit** suite (MotionSense · SpeakSense · HearSense · VisionSense). Screening & self-monitoring only — not a medical diagnosis. Phase 2 = native app with calibrated sensors.